Exam :	End Term Quiz
Subject :	AI
Total Marks :	25.00
QP :	2024 Dec22: IIT M FN EXAM QDB2

Exam Mode Learning Mode

View Question Paper Summary

QUESTION MENU

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34

TIMER

00:17

CONTROLS

SUBMIT EXAM

Your Score

0.00 / 25.00

(0%)

Question 1 : 6406531041196

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT **"DEGREE LEVEL : AI: SEARCH METHODS FOR PROBLEM SOLVING (COMPUTER BASED EXAM)" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL**

TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

OPTIONS :

- YES
- NO

Your score : 0

Discussions (0)

Question 2 : 6406531041197

Total Mark : 0.00 | Type : MSQ

OPTIONS :

- [ ] Printed graph sheets were provided on time.
- [ ] Printed graph sheets were provided late.
- [ ] Printed graph sheets were not provided.
- [ ] I used the graph sheets.
- [ ] I did not use graph sheets.

Your score : 0

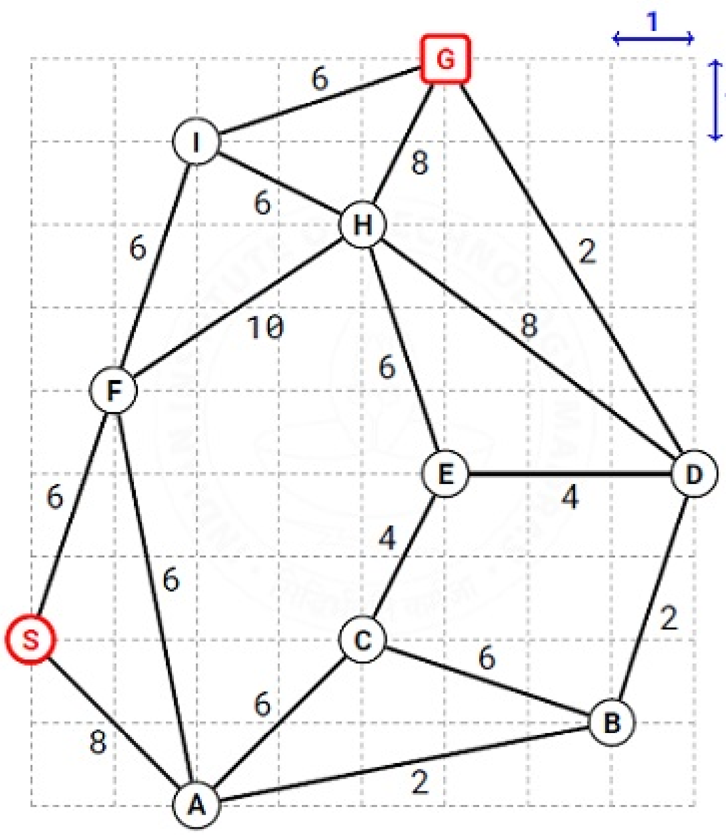
Discussions (0)

Question 3 : 6406531041198

Total Mark : 0.00 | Type : COMPREHENSION

SEARCH

The figure shows a map on a uniform grid where each tile is 1x1 in size. The start node is S and the goal node is G. The MoveGen function returns nodes in alphabetical order. Use Manhattan Distance as the heuristic function. **Tie-breaker:** If several nodes have the same cost, use node labels to break the tie.
Based on the above data, answer the given subquestions.



Your score : 0

Question 4 :
6406531041199

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

What is the path found by the Best First Search algorithm? Enter the path as a comma separated list of node labels.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: S,X,Y,Z,G**

Answer (Alphanumeric):

Answer

Accepted Answer : S,F,H,G

Your score : 0

Discussions (0)

Question 5 : 6406531041200

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

What is the path found by A* search algorithm? Enter the path as a comma separated list of node labels. NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: S,X,Y,Z,G**

Answer (Alphanumeric):

Answer

Accepted Answer : S,F,I,G

Your score : 0

Discussions (0)

Question 6 : 6406531041201

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

What is the path found by Branch-and-Bound search algorithm? Enter the path as a comma separated list of node labels.
Use the Branch-and-Bound variation that avoids cyclic expansions like S,A,S,A,S,A,...
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: S,X,Y,Z,G**

Answer (Alphanumeric):

Answer

Accepted Answer : S,A,B,D,G

Your score : 0

Discussions (0)

Question 7 :
6406531041202

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : MCQ

For the given map, which algorithm finds the shortest path from S to G?

OPTIONS :

- Best First Search
- A* Search Algorithm
- Branch-and-Bound Search Algorithm
- None of these.

Your score : 0

Discussions (0)

Question 8 :
6406531041203

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : MCQ

What can you say about the heuristic function for the given graph?

OPTIONS :

- Admissible
- Inadmissible
- Partly admissible and partly inadmissible
- Cannot be determined

Your score : 0

Discussions (0)

Question 9 : 6406531041204

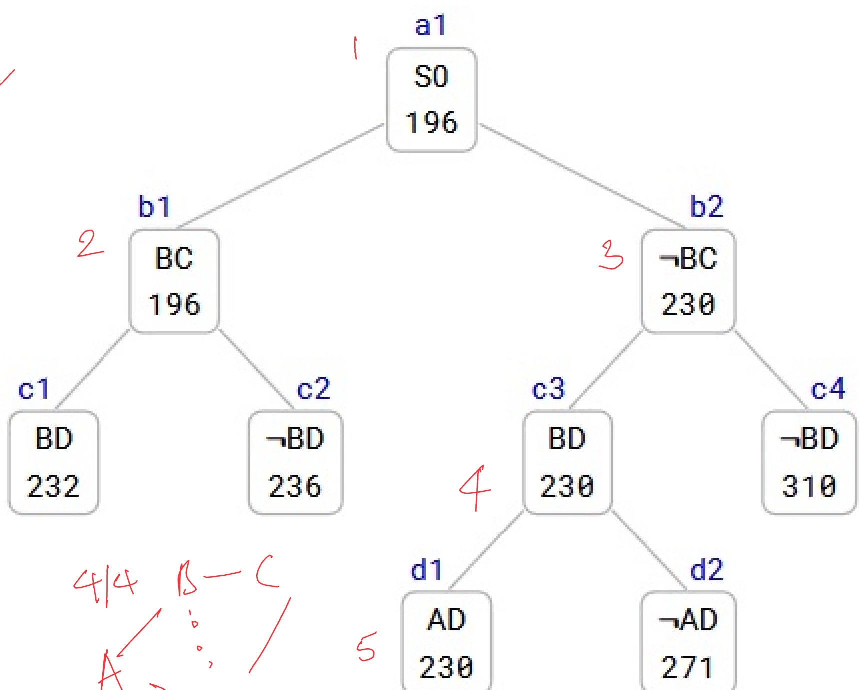
Total Mark : 0.00 | Type : COMPREHENSION

TSP Branch-and-Bound

The TSP Branch-and-Bound algorithm is solving a TSP instance where the cities are A, B, C, ... and so on. The Branch-and-Bound search tree at the time when **the algorithm has discovered the optimal tour** is shown below.

Each node in the search tree displays an edge (either XY or ~XY), a cost value, and a unique reference number (a1, b1, b2, c1, c2, c3, c4, d1, d2). Use the reference numbers to break ties. When required, enter the reference numbers in short answers.

What information can you glean from the search tree? Answer the sub-questions based on the information gleaned from the search tree.



Your score : 0

Question 10 : 6406531041205

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : SA

Let S0 (ref. no. a1) be the first node to be refined, identify the next three nodes (2nd, 3rd and 4th node) refined by the TSP Branch-and-Bound algorithm. Enter the nodes (node reference numbers) in the order they are refined.
Enter a comma separated list of node reference numbers. NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: b9,c9,d9**

Answer (Alphanumeric):

Answer

Accepted Answer : b1,b2,c3

Your score : 0

Discussions (0)

Question 11 : 6406531041206

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Which node represents the optimal tour and what is the cost of the optimal tour? Enter the node reference number and the tour cost in the text box, or enter NIL if it is not possible to determine the optimal tour.
Enter a node reference number followed by tour cost, separated by comma. NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: a9,42**

Answer (Alphanumeric):

Answer

Accepted Answer : d1,230

Your score : 0

Discussions (0)

Question 12 : 6406531041207

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Determine the number of cities in the TSP instance. Enter the number of cities in the text box, or enter NIL if it is not possible to determine the number of cities.
Enter an integer. NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: 42**

Answer (Numeric):

Answer

Accepted Answer : 5

Your score : 0

Discussions (0)

Question 13 : 6406531041208

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Start from city A, what is the path representation of the optimal tour? Enter the path representation in the text box, or enter NIL if it is not possible to determine the optimal tour.
Enter a comma separated list of cities (city labels). NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: A,X,Y**

Answer (Alphanumeric):

Answer

Accepted Answer : A,C,E,B,D

Your score : 0

Discussions (0)

Question 14 : 6406531041209

Total Mark : 0.00 | Type : COMPREHENSION

GAMES

The figure shows a game tree with evaluation function values at the leaf nodes. The

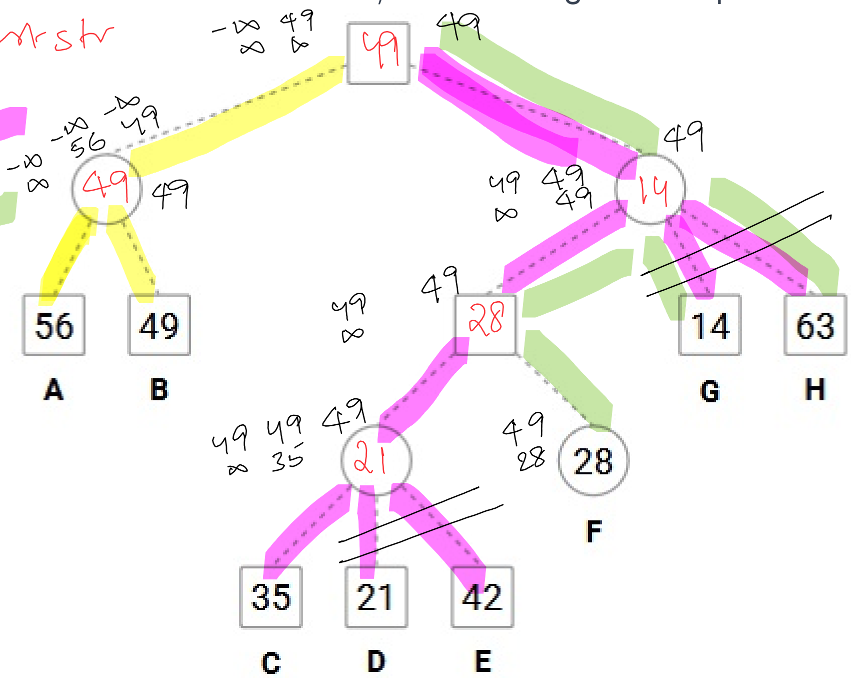
leaf nodes are labeled from A to H. Use these labels (A to H) in short answers.
Based on the above data, answer the given subquestions.

max strategy

A,B → best str

CDEGH

FGH



Your score : 0

Question 15 :
6406531041210

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : MSQ

Which of the following is a strategy for the MAX player?

OPTIONS :

- [ ] A,C,F
- [ ] C,D,E,G,H
- [ ] F,G,H
- [ ] G,H

Your score : 0

Discussions (0)

Question 16 : 6406531041211

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

List the leaf nodes in the best strategy for MAX, and enter the label(s) of those leaf nodes in alphabetical order.
Enter a comma separated list of node labels in alphabetical order. NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: X,Y,Z**

Answer (Alphanumeric):

Answer

Accepted Answer : A,B

Your score : 0

Discussions (0)

Question 17 : 6406531041212

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Identify all the leaf nodes that do not affect the value of the game when played perfectly, and enter the label(s) of those leaf nodes in alphabetical order.
Enter a comma separated list of node labels in alphabetical order. NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: X,Y,Z**

Answer (Alphanumeric):

Answer

Accepted Answer : D,E,G,H

Your score : 0

Discussions (0)

Question 18 : 6406531041213

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

MAX priority Que

List the leaf nodes solved (assigned SOLVED status) by SSS*, and enter the labels (A to H) of those leaf nodes in alphabetical order.

Tie-breaker: when several nodes carry the same best cost then select the deepest node, if tie persists then select the leftmost of the deepest nodes.

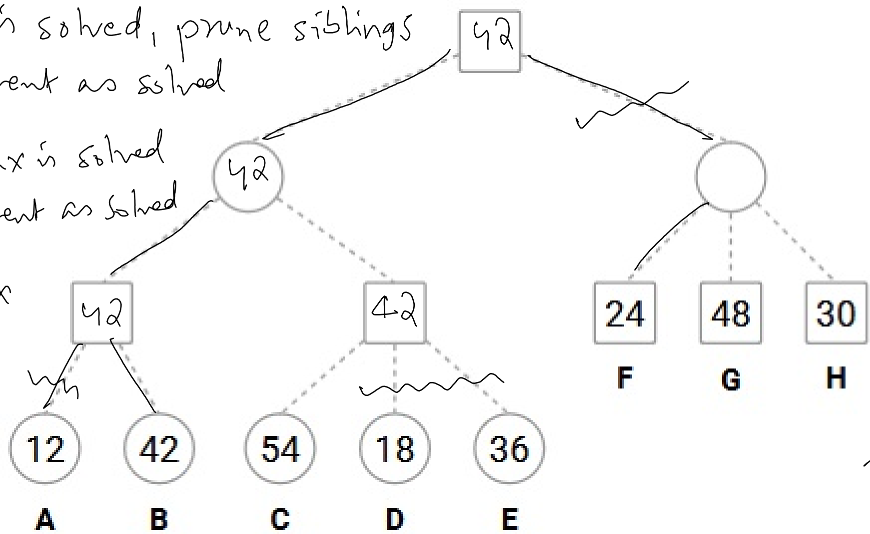
Enter a comma separated list of node labels in alphabetical order. NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: X,Y,Z**

when min is solved, prune siblings
label parent as solved

when last max is solved
label parent as solved

when mid max
is solved
go to sibling

when leaf is
reached
mark solved
and
take min



A,S,12, B,S,42, F,S,24
C,S,42
C,L,42
C,L,42 D,L,42 E,L,42
C,S,42
B,S,42

$\boxed{\text{A B C F}}$

Answer (Alphanumeric):

Answer

Accepted Answer : A,B,C,F

Your score : 0

Discussions (0)

Question 19 : 6406531041222 Total Mark : 0.00 | Type : COMPREHENSION

AUTOMATED PLANNING

The domain description of a Blocks World with a single one-armed robot is given below. This is the same domain used in the assignments.

Consider the planning problem with the following start state and goal description.

PREDICATES

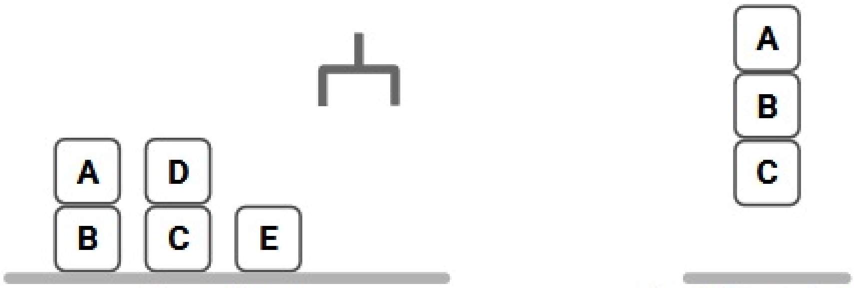
```
armEmpty        The arm is not holding any block, it is empty.
holding(X)      The arm is holding X.
onTable(X)      X is on the table.
clear(X)        X has nothing above it, it is clear.
on(X,Y)         X is directly placed on Y.
```

OPERATORS

```
Pickup(X): pick up X from the table.

     Preconditions: { armEmpty, clear(X), onTable(X) }
     Add Effects  : { holding(X) }
     Del Effects  : { armEmpty, onTable(X) }
Putdown(X): place X on the table.

     Preconditions: { holding(X) }
     Add Effects  : { armEmpty, onTable(X) }
     Del Effects  : { holding(X) }

Unstack(X,Y): pick up X that is directly sitting on Y.

     Preconditions: { armEmpty, clear(X), on(X,Y) }
     Add Effects  : { clear(Y), holding(X) }
     Del Effects  : { armempty, on(X,Y) }

Stack(X,Y): place X directly on top of Y.

     Preconditions: { holding(X), clear(Y) }
     Add Effects  : { armEmpty, on(X,Y) }
     Del Effects  : { holding(X), clear(Y) }
```

AUTOMATED PLANNING

The domain description of a Blocks World with a single one-armed robot is given below. This is the same domain used in the assignments.
Consider the planning problem with the following start state and goal description.
Based on the above data, answer the given subquestions.



Start State

{ armEmpty,
clear(A), clear(D), clear(E),
on(A,B), on(D,C),
onTable(B), onTable(C), onTable(E) }

Goal Description

{ on(B,C), on(A,B) }

Your score : 0

Question 20 : 6406531041223

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : MSQ

Which of the following are **applicable** actions in the start state?

OPTIONS :

- [ ] Pickup(A)
- [ ] Pickup(E)
- [ ] Stack(A,B)
- [ ] Stack(B,C)
- [ ] Unstack(A,B)
- [ ] Unstack(B,A)
- [ ] Unstack(D,C)

Your score : 0

Discussions (0)

Question 21 : 6406531041224

View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : MSQ

Which of the following are **relevant** actions in the goal state?

OPTIONS :

- [ ] `Pickup(A)`
- [ ] `Pickup(E)`
- [ ] `Stack(A,B)`
- [ ] `Stack(B,C)`
- [ ] `Unstack(A,B)`
- [ ] `Unstack(B,A)`
- [ ] `Unstack(D,C)`

Your score : 0

Discussions (0)

Question 22 : 6406531041225

View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : MSQ

In the planning graph, which of the following are mutex action pairs in Layer 1?

OPTIONS :

- [ ] `Pickup(E), NO-OP for armEmpty`

- [ ] `Pickup(E), NO-OP for clear(E)`
- [ ] `Pickup(E), Unstack(A,B)`
- [ ] `Unstack(A,B), NO-OP for on(D,C)`
- [ ] `Unstack(A,B), Unstack(D,C)`

Your score : 0

Discussions (0)

Question 23 : 6406531041226

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : MSQ

In the planning graph, which of the following are mutex proposition pairs in Layer 1?

OPTIONS :

- [ ] `clear(B), armEmpty`
- [ ] `clear(E), clear(B)`
- [ ] `clear(E), on(A,B)`
- [ ] `holding(A), holding(D)`

Your score : 0

Discussions (0)

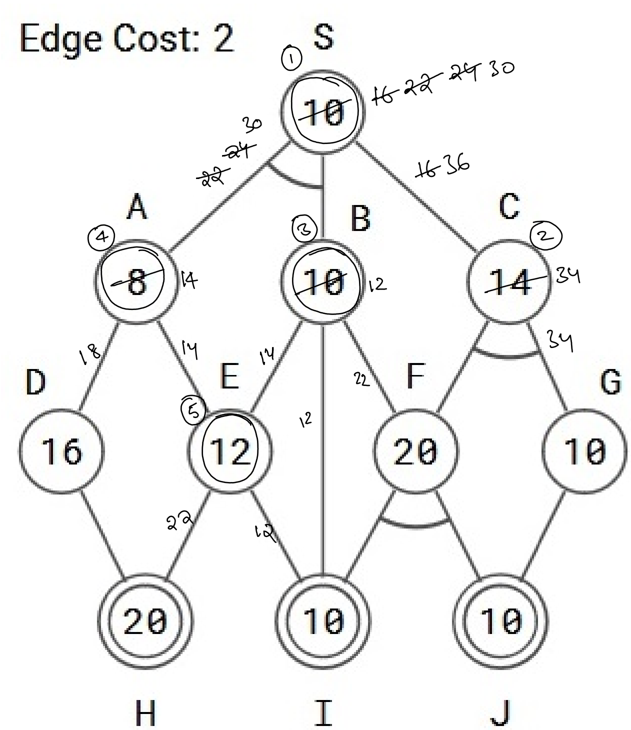
Question 24 : 6406531041214

Total Mark : 0.00 | Type : COMPREHENSION

PROBLEM DECOMPOSITION

The figure shows an AND-OR graph that depicts how a problem S can be decomposed into one or more smaller problems. Nodes are uniquely identified by

labels (S, A, B, C, ...). The number in each node is the heuristic estimate of the cost of solving that node.
Nodes shown in double lines are primitive nodes and their values are actual costs. Observe that a primitive node is added to the graph by its parent when the parent is expanded, and the primitive node is labeled as SOLVED and it will not be expanded subsequently.
The cost of each edge is 2 units.
Tie-breaker 1: If several nodes have the same cost then break the tie using node labels. **Tie-breaker 2:** For AND nodes, select the unsolved branch having the highest cost.
Use AO* algorithm to solve S, then answer the sub-questions.



Your score : 0

Question 25 : 6406531041215

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : SA

List the first three nodes (including S) expanded by AO* algorithm. List the nodes in the order they are expanded. Observe that primitive nodes are not expanded.
Enter a comma separated list of node labels. NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: X,Y,Z**

Answer (Alphanumeric):

Answer

Accepted Answer : S,C,B

Your score : 0

Discussions (0)

Question 26 : 6406531041216

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Determine the value of the start node S after each node is expanded. What are the values of S after the 1st, 2nd and 3rd nodes are expanded, respectively? Enter the 3 values in the textbox. Enter a comma separated list of numbers. NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: 12,42,17**

Answer (Alphanumeric):

Answer

Accepted Answer : 16,22,24

Your score : 0

Discussions (0)

Question 27 : 6406531041217

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

What is the final value of the start node S?
Enter a number. NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: 42**

Answer (Numeric):

Answer

Accepted Answer : 30

Your score : 0

Discussions (0)

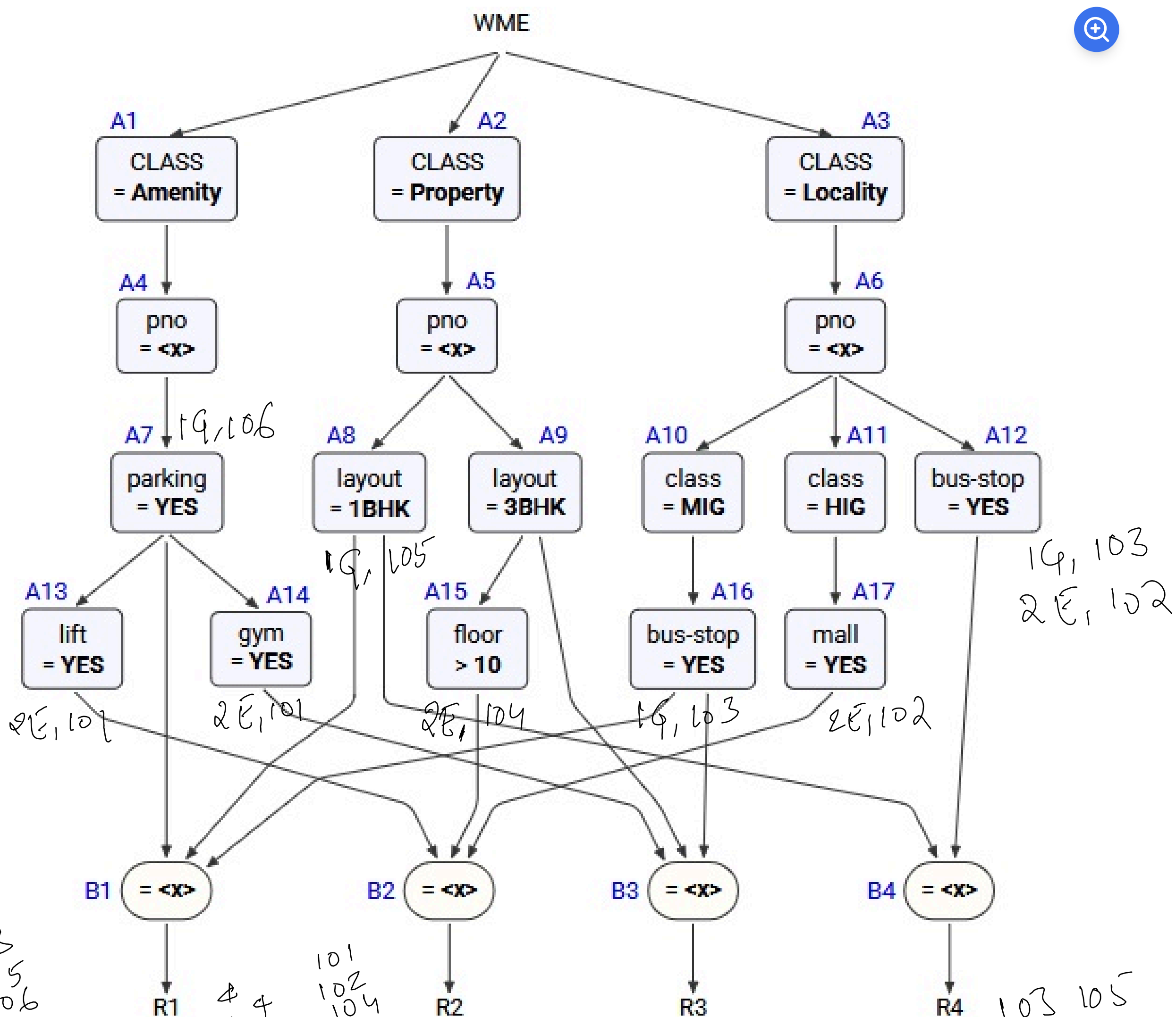
Question 28 : 6406531041218

Total Mark : 0.00 | Type : COMPREHENSION

RULE BASED EXPERT SYSTEMS

A Rete Net for classification of properties is shown in the figure. The labels A1, A2, A3, ..., A10, A11, A12, A13, ..., and B1, B2, B3, B4 uniquely identify nodes in the network. When required, use the above label ordering to **break ties** and to enter short answers.

Run the Rete algorithm for the Working Memory shown below, the WMEs are in timestamp order. Assume that WMEs reside at appropriate Alpha nodes, and the Beta nodes point to WMEs residing in Alpha nodes.



4 +3 +3

10

103 105 106

4 + 4 + 4

101 102 104

12

103 105

3 + 3 = 6

RULE BASED EXPERT SYSTEMS

A Rete Net for classification of properties is shown in the figure. The labels A1, A2, A3, ..., A10, A11, A12, A13, ..., and B1, B2, B3, B4 uniquely identify nodes in the network. When required, use the above label ordering to **break ties** and to enter short answers.

Run the Rete algorithm for the Working Memory shown below, the WMEs are in timestamp order. Assume that WMEs reside at appropriate Alpha nodes, and the Beta nodes point to WMEs residing in Alpha nodes.

For each WME identify its location (node label) in the Rete Net, and prepare the conflict set for the first cycle, then answer the sub-questions.

```
101. (Amenity   ^pno 2E  ^parking YES  ^lift YES  ^gym YES)
102. (Locality  ^pno 2E  ^class HIG  ^mall YES  ^bus-stop YES)
103. (Locality  ^pno 1G  ^class MIG  ^bus-stop YES)
104. (Property  ^pno 2E  ^layout 3BHK  ^floor 44)
105. (Property  ^pno 1G  ^layout 1BHK  ^floor 1)
106. (Amenity   ^pno 1G  ^parking YES  ^heating YES)
```

Your score : 0

Question 29 : 6406531041219

View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : MSQ

Which of the following rule-data tuples are in the conflict-set?

OPTIONS :

- [ ] R1,103,105,106
- [ ] R2,101,102,104
- [ ] R3,101,103,106
- [ ] R4,103,105

Your score : 0

Discussions (0)

Question 30 :
6406531041220

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : MCQ

If the Inference Engine uses **Specificity** as the conflict resolution strategy then which of the following rule-data tuples will qualify?

OPTIONS :

- R1,103,105,106
- R2,101,102,104
- R3,101,103,106
- R4,103,105

Your score : 0

Discussions (0)

Question 31 :
6406531041221

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : MCQ

If the Inference Engine uses **Recency** as the conflict resolution strategy then which of the following rule-data tuples will qualify?.

OPTIONS :

- R1,103,105,106
- R2,101,102,104
- R3,101,103,106

○ R4,103,105

Your score : 0

Discussions (0)

Question 32 : 6406531041227 Total Mark : 0.00 | Type : COMPREHENSION

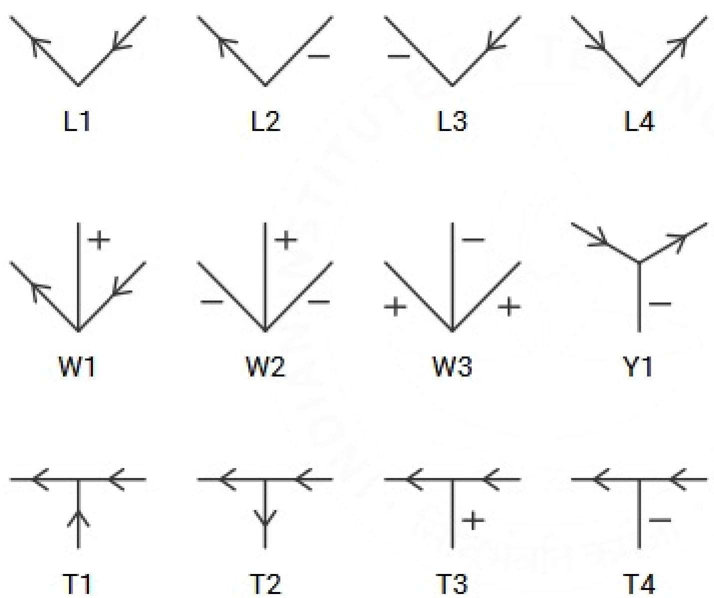
CONSTRAINT SATISFACTION

The set of junctions (L, W, Y and T type junctions) that occur in a 2D line drawing of trihedral objects is provided below. The in-plane clockwise/counterclockwise rotations of these junctions are valid as well. These junctions provide constraints on the possible edge assignments (convex, concave, arrow) for the edges/lines in 2D line drawings of trihedral objects.

The junctions carry unique labels: L1, L2, L3, L4, L5, L6, T1, T2, T3, T4, W1, W2, W3, Y1, Y2, Y3. When required, use the labels in short answers.

Note: A 2D line drawing of trihedral objects is considered to be consistent if all the edges and junctions can be assigned labels that are consistent with each other, otherwise the drawing is considered to be inconsistent and all labels are reset to NIL.

Apply a suitable algorithm to assign consistent labels to edges/junctions in the 2D line drawings in the sub-questions. Choose a suitable edge and junction order for solving the problems.



Your score : 0

Question 33 : 6406531041228

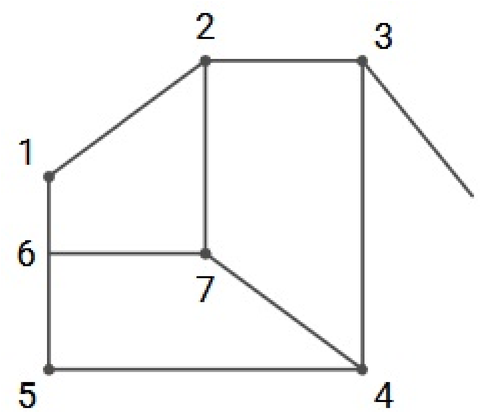
View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : SA

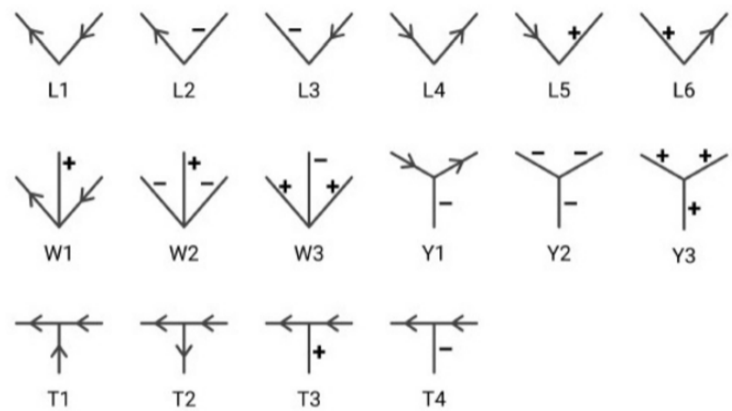
T,W,Y L

Assign consistent labels to all the edges and junctions in the 2D line drawing shown below. Enter the labels of the junctions 1, 2, 3, 4 in the text box, in that order. Enter NIL if the drawing has no consistent label assignment.

Enter a comma separated list of junction labels, or enter NIL. NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: Y9,T9,W9,Y9**



L



Answer (Alphanumeric):

Answer

Accepted Answer : NIL

Your score : 0

Discussions (0)

Question 34 : 6406531041229

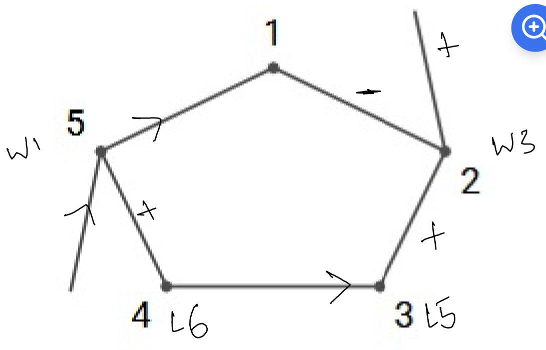
View Parent QN | View Solutions (0)

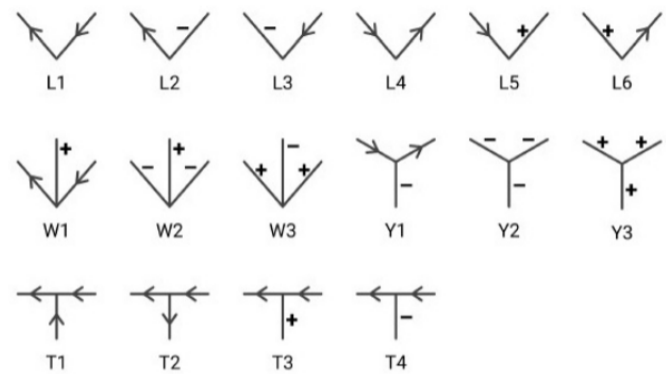
Total Mark : 1.00 | Type : SA

Assign consistent labels to all the edges and junctions in the 2D line drawing shown below. Enter the labels of the junctions 1, 2, 3, 4 in the text box, in that order. Enter NIL if the drawing has no consistent label assignment.

Enter a comma separated list of junction labels, or enter NIL. NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS. **Answer format: Y9,T9,W9,Y9**

TWY





Answer (Alphanumeric):

Answer

Accepted Answer : L3,W3,L5,L6

Your score : 0

Discussions (0)

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